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Assignment-3

Cyber Security – Malware and Threat Detection

Module 3: CS - Cyber threats & CEH

1. What are the different types of hacking methods?

ANS: - Hacking methods are techniques used by attackers to gain unauthorized access to systems.

- Phishing: - fake emails/websites used to steal login credentials.
- Malware Attacks: - Malicious software like virus, worm, trojans, Ransomware.
- Password Attacks: - trying to guess or crack passwords.
- Man-in-the-Middle (MITM): - Attacker intercepts communication between two users.
- SQL Injection: - Injecting malicious SQL queries into databases and used to steal or modify data.
- Denial-of-Service (Dos/DDos): - Flooding server with traffic – make it unavailable.

2. Explain Types of Password Attacks

ANS: - Password attacks are used to gain access to accounts.

- Brute Force Attack: - tries all possible combinations, Slow but guaranteed.
- Dictionary Attack: - Uses a list of common passwords, faster than brute force.
- Hybrid Attack: - Combines dictionary + brute force
- Credential Stuffing: - Uses leaked username/password from other sites.
- Keylogging: - Records keystrokes to capture passwords.

3. Explain Password Cracking Tools: pwdump7, Medusa and Hydra

ANS: - pwdump7: - Used to extract password hashes from Windows, works with SAM database, Used with cracking tools like john the Ripper.

- Medusa: - Fast network login brute-force tool, it supports SSH, FTP, HTTP, SMB.
- Hydra: - Very popular password cracker, support many protocols: SSH, FTP, Telnet, HTTP, RDP.

4. Explain Types of Steganography with QuickStego and Echo

ANS: - It means hiding data inside another file (image/audio/video/text).

- Audio Steganography: - Data hidden in sound files.
- Image Steganography: - Data hidden in image pixels, uses LSB (Least Significant Bit).
- Video Steganography: - Combines image + audio techniques.
- Text Steganography: - Uses spaces, fonts, or hidden characters.

5. Perform Practical on key logger tool.

ANS: - A keylogger records keystrokes.

- Basic Working: - User install/executes keylogger → it runs in background → Captures keystrokes → Stores or sends data to attacker.

• Malware

1. Define Types of Viruses.

ANS: - File Infector Virus: - Infects executable files (.exe, .com), Spreads when the file runs.

- Boot Sector Virus: - Infects Master Boot Record (MBR), Loads before OS starts.
- Macro Virus: - written in macro language (Eg: - word, excel), spreads via documents.
- Resident Virus: - lives in RAM, Infects files automatically.
- Multipartite Virus: - Infect multiple parts (files + boot sector).

2. Create viruses using Http Rat Trojan tool.

ANS: - RAT = Remote Access Trojan, gives attacker remote control over victim system.

- How HTTP RAT works: - Attackers create malicious payload → Victim executes file → Connection established to attacker server (C2) → Attacker control system remotely: - file access, screen monitoring, Keylogging.

3. Explain any one Antivirus with example.

ANS: - Antivirus is the software that detects, prevents, and removes malware.

- Quick Heal Total Security: - features of quick heal are → Real-time protection, Malware detection (virus,

spyware, ransomware), web security (blocks malicious websites), Email protection, Firewall.

- How Antivirus Works: - Signature-based detection (matches known virus patterns)
    - Heuristic analysis: - Detects unknown threats by behavior.
    - Behavior monitoring: - Tracks suspicious activity.